Ang Zhe Hao (Nicholas)

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EDUCATION

Singapore Management University *Aug 2024 - May 2028*

- BSc (Software Engineering), second major in Software Systems
- Cumulative Grade Point Average (CGPA): 3.37/4.00
- Coursework: Advanced Databases, Operating Systems, Cyber-Physical Systems, Generative AI with LLMs

TECHNICAL SKILLS

- Languages: C++, Python, Java, C, Typescript, Javascript, PHP, SQL, HTML/CSS
- Frameworks & Libraries: CMake, React.js, Node.js, Express.js, FastAPI, Spring Boot, Vue.js, Tailwind CSS
- Databases & Messaging: DynamoDB, RabbitMQ, MySQL, MQTT, AMQP, PostgreSQL
- DevOps & Tools: Docker, Docker Compose, Kubernetes, AWS, Azure, Jira, Git, Linux, Grafana, Terraform, GitLab
- Certifications: AWS Certified Solutions Architect - Associate, AWS Certified Developer - Associate

Experience

Systems Engineer Intern | Terraform, Java, Springboot, Kubernetes, AWS, Azure *May 2026 - Dec 2026*
GovTech Open Digital Platform (ODP)

- Maintained and extended Terraform modules across a multi-service production stack - including identity, secrets, context brokering and messaging infrastructure - to meet security compliance requirements
- Independently designed an Azure-based production architecture mirroring existing AWS infrastructure to support cloud migration planning and cost evaluation
- Enhanced CI/CD security by migrating autogenerated source-code artifacts from agency-owned repositories to a controlled internal platform, reducing unauthorized access exposure

Wind Turbine Monitoring System - Industry Project (EcoEdge) | Coursework | Python, C++, MQTT *Aug 2025 - Nov 2025*

- Engineered end-to-end IoT system integrating Raspberry Pi gateway with Arduino actuator for wind turbine simulation
- Developed Python services on Raspberry Pi and programmed Arduino in C++ to process Hall Effect sensor readings with sub-second telemetry latency and deterministic actuator response

RELEVANT PROJECTS

Order Book Implementations | C++, CMake | **GitHub** *May 2026 - Jun 2026*

- Implemented and benchmarked four order book data structures (map, unordered_map, vector, priority queue) across 20 market scenarios spanning normal, spike, trending, HFT, and random conditions
- Achieved ~1.2M orders/sec with a map-based implementation, outperforming unordered_map by 1.5x and vector by 2x, driven by std::map's ordered traversal enabling O(1) best-bid/ask retrieval versus O(n) scans
- Built a CMake-integrated benchmark suite using Google Benchmark, measuring per-order latency and throughput with statistically stable results across implementations

TCPH Specialized Preprocessor | Coursework | Python, C++, Arrow, Parquet, DuckDB **| GitHub** *Jan 2026 - May 2026*

- Designed a specialized processor for TPC-H Q13 over Parquet datasets, achieving up to 2.28× faster execution than DuckDB by replacing generic join-and-aggregate with a single-pass counting strategy
- Engineered Python and C++ implementations using Apache Arrow, eliminating join intermediates and reducing aggregation overhead while maintaining exact query correctness
- Established benchmarking and profiling tools to analyse performance across TPC-H scale factors up to SF5, demonstrating consistent 2x+ speedups on analytical workloads

Quant Trading Bot SG vs HK Hackathon | Python, AWS, Termux **| GitHub** *Feb 2026 - Mar 2026*

- Led a team to deploy a trading bot with a trading strategy on a simulated cryptocurrency environment
- Reached a peak of 1.4% profits using live Binance API data over 1 week, attaining a peak of 4th out of 50+ SG teams, by developing a trading bot that ingests Binance API historical data and runs on AWS EC2

LEADERSHIP & VOLUNTEERING

President | SMU Strategica Board Games Club *Jan 2026 – Ongoing*

- Directed 8-member executive committee managing weekly operations for 70+ active members

Volunteer | SMU Project Kidleidoscope *Jul 2024 – Dec 2024*

- Facilitated science and arts & crafts activities for around 30 children aged 7–12 at various childcare centres